Surface Texture

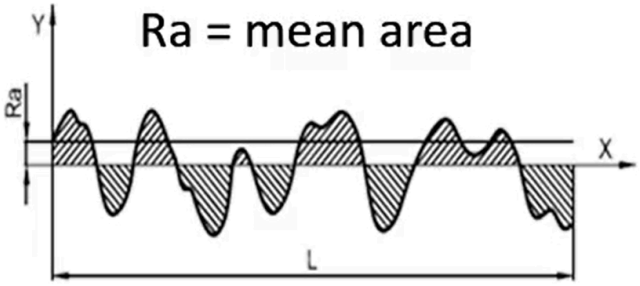
Surface refers to the surface finish of a part, which has to be controlled to reduce friction and wear of parts. The appropriate surface finish used is determined by the parts function, nominal size, loading types, operating speed and conditions, mating surfaces & characteristics, and the type/amount of lubricant.

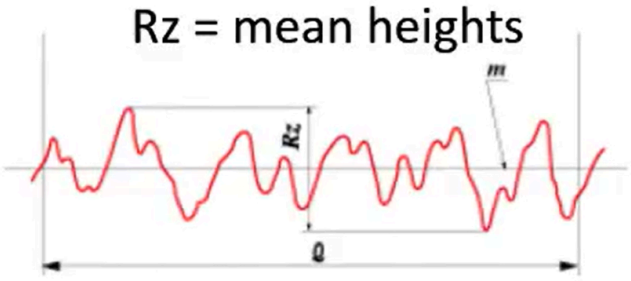
Surface Texture Metrics

Typically, we use R_a and R_z to represent the roughness of a surface in micrometers.

R_a is the sum of the profile areas above and below a centerline from a sample length, divided by the length of the sample length, essentially the mean profile area.

R_z is the sum of the profile's peak and valley heights above and below a centerline from a sample length, divided by the length of the sample length, giving the mean height.





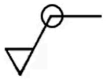
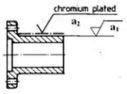
The two values can be converted with conversion tables.

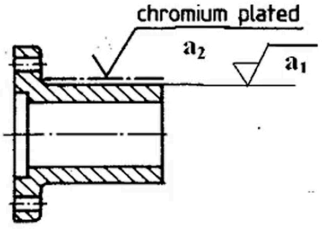
Depending on the maximum surface roughness allowed, different manufacturing processed may not be permitted as they may not be able to reach the finish required:

Process	Ra (μm)
Flame cutting	25 – 12.5
Sawing	25 – 1.6
Planing, Shaping	12.5 – 1.6
Drilling	6.3 – 1.6
Milling	6.3 – 0.80
Reaming	3.2 – 0.80
Boring, Turning	6.3 – 0.40
Grinding	1.6 – 0.10
Honing	0.80 – 0.10
Polishing	0.40 – 0.10
Lapping	0.40 – 0.05

Symbol Indication

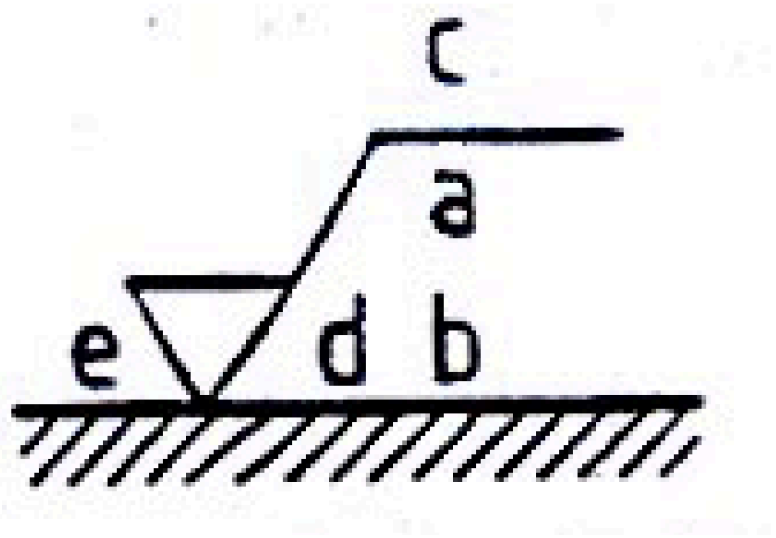
On drawings, we specify the surface roughness of a surface with the following symbols.

Symbol	Meaning
	Basic symbol. Needs to be accompanied by comments or Specifications e.g. 'to be machined if necessary'.
	Removal of material is required.
	Removal of material is not permitted.
	Same surface texture is required on <u>all</u> surfaces around a workpiece outline.
	Specification of surface treatment/coating a1 = roughness before coating a2 = roughness after coating



These symbols only indicate how the surface roughness should be achieved, not the actual roughness metric.

We assign comments to these symbols like so, in the full ISO symbol:



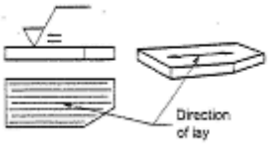
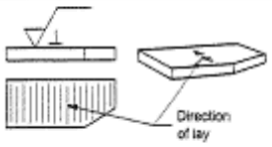
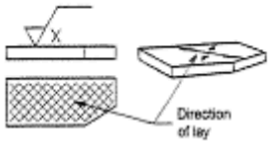
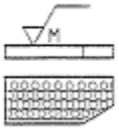
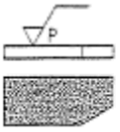
a: max roughness value

b: additional texture surface requirements

c: manufacturing method to be used, in past tense, small letters (e.g. reamed)

d: surface lay; the direction of the surface finish

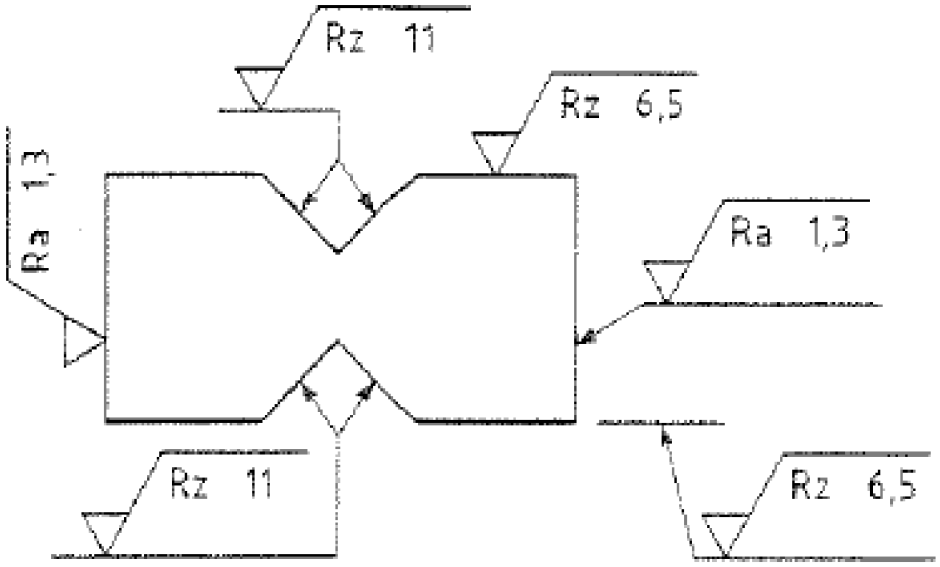
the surface lay symbol is from this set:

Graphical symbol	Interpretation and example	
=	Parallel to plane of projection of view in which symbol is used	
⊥	Perpendicular to plane of projection of view in which symbol is used	
X	Crossed in two oblique directions relative to plane of projection of view in which symbol is used	
M	Multi-directional	
C	Approx. circular relative to centre of surface to which symbol applies	
R	Approx. radial relative to centre of surface to which symbol applies	
P	Lay is particulate, non-directional, or protuberant	

If it is necessary to specify a surface pattern which is not clearly defined by these symbols, this shall be achieved by the addition of a suitable note to the drawing.

e: machining allowance

Shown is an example of how to use these symbols:



Some things to note:

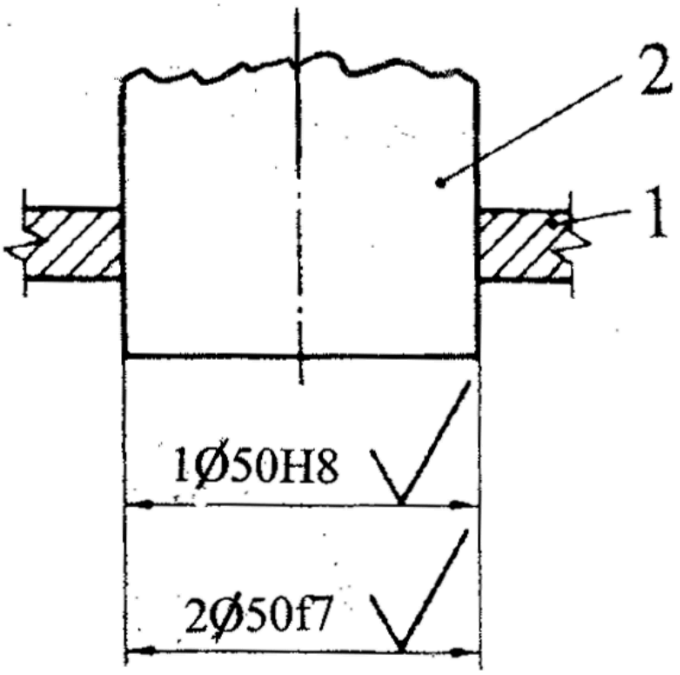
all comments are either upright or leftward, as ISO 129 specifies that inscriptions must be readable from either the bottom or the right side of the drawing.

Typically, surface texture symbols are applied directly on the part's outline itself, but may be applied on extension lines or arrow lines.

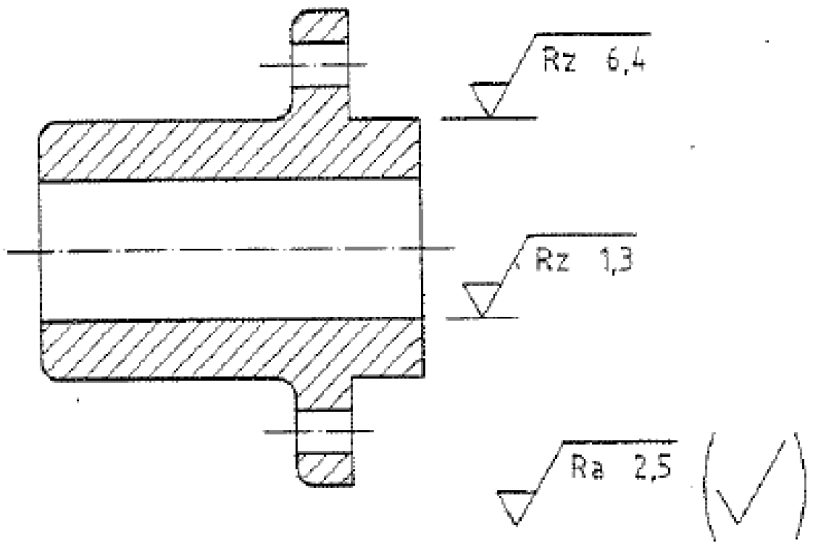
Extension lines, arrow lines and the surface texture symbols all lie outside the part drawing too.

There are also some special cases to using surface texture symbols:

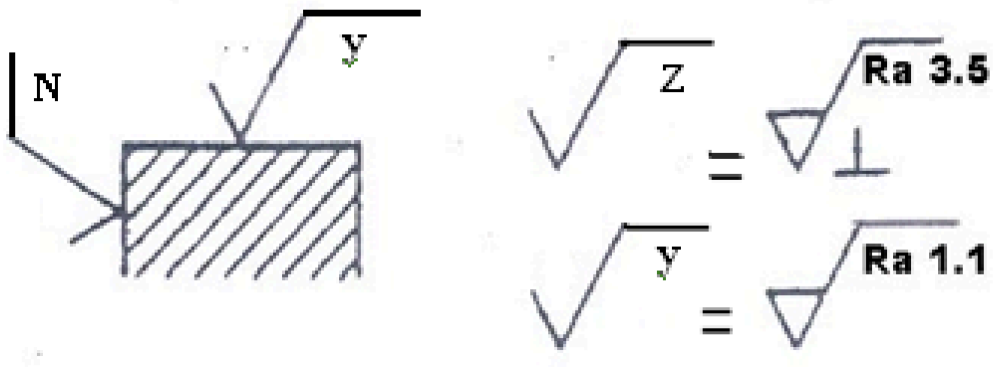
In the case of assembly drawings, surface texture symbols may be applied to the dimensions itself:



If majority of the surfaces on a part needs to be a certain texture, one symbol placed outside but adjacent to the part drawing, with a basic symbol in brackets can represent that. Any exceptions to the majority are notated as usual, like in this case the interior must have a max surface roughness of Rz 1.3.



If a surface finish is used multiple times, a short hand may be used to avoid clutter.



By applying these surface texture symbols, it becomes another metric that needs to be maintained and controlled by the machinist, hence only use it when it serves an actual purpose, and avoid using it if the manufacturing process used already achieves the desired surface finish.